

Curriculum Vitae - Alexander Abuabara

1 Profile

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2 Education

- **Ph.D. in Urban and Regional Science**, Texas A&M University, College Station (2022)
Dissertation: Applications of Causal Bayesian Networks in Urban Planning
Committee: Walter Gillis Peacock (chair), Zenon Medina-Cetina (chair), Douglas F. Wunneburger, David H. Bierling
- **Doctoral Studies in Production Engineering**, Federal University of S. Carlos, Brazil (2007-2009)
Advisor: Reinaldo Morabito Neto
- **Master of Science in Production Engineering**, Federal University of São Carlos, Brazil (2006)
Thesis: A Mixed Integer Programming Approach Applied to Multi-Period Industrial Planning (original in Portuguese)
Committee: Reinaldo Morabito Neto (chair), Horacio Hideki Yanasse, Marcos Nereu Arenales, Denise Sato Yamashita
- **Bachelor of Science in Civil Engineering**, University of São Paulo, Brazil (2001)

3 Experience

3.1 Academic Positions

- **Instructional Assistant Professor** (Fall 2025-present)
Wm Michael Barnes '64 Department of Industrial and Systems Engineering, Texas A&M University
Supervisor: Satish Bukkapatnam, Lewis Ntamo
- **Instructional Assistant Professor** (Fall 2024-Spring 2025)
Department of Landscape Architecture and Urban Planning, Texas A&M University
Supervisor: Galen D. Newman
- **Postdoctoral Research Associate** (Fall 2022-Spring 2024)
Department of Landscape Architecture and Urban Planning, Texas A&M University
Supervisors: Walter Gillis Peacock, Galen D. Newman
- **Graduate Research Lecturer** (Fall 2019-Spring 2022)
Department of Landscape Architecture and Urban Planning, Texas A&M University
Supervisor: Galen D. Newman, Shannon Van Zandt
- **Graduate Research Assistant** (Spring 2015-Spring 2022)
Department of Landscape Architecture and Urban Planning, Texas A&M University
Supervisor: Walter Gillis Peacock

3.2 Industry Experience

- **Civil Engineer (2002-2005)**
Department of Research, Sucocitrico Cutrale LLC, Araraquara, Brazil
Supervisor: Marcos Neves Penteado Moraes
- **Civil Engineer Trainee (1999-2000)**
PRO-Salas, University of São Paulo, São Carlos, Brazil
Supervisor: Manoel Rodrigues Alves

4 Teaching

4.1 Courses Taught

Industrial and Systems Engineering, Texas A&M University

(Average Teaching Evaluation 4.63/5)

- **DAEN 301 Data Engineering Coding Experience II (Spring 2027)**
Credit 1. 0 Lecture Hours. 3 Lab Hours. Application of computational tools to model and solve data engineering problems involving stochastic systems, reinforcement learning, ensemble learning and data visualization.
- **DAEN 300 Data Engineering Coding Experience I (Fall 2026)**
Credit 1. 0 Lecture Hours. 3 Lab Hours. Application of computational tools to model and solve data engineering problems primarily involving machine learning and optimization techniques.
- **DAEN 489 / ISEN 489 Spatial Data Science in Engineering (Spring 2026)**
Credits 3. 3 Lecture Hours. This course provides engineers with both theoretical foundations and practical skills in spatial databases and geographic information systems (GIS), with emphasis on spatial data modeling, storage, querying, analysis, and visualization. Students will learn to use open source tools to handle real spatial datasets, build spatial databases, perform spatial analysis, and integrate GIS solutions in engineering workflows.
- **DAEN 427 / ISEN 427 Decision and Risk Analysis (Fall 2025)**
Credits 3. 3 Lecture Hours. Overview of the state of the art in descriptive and prescriptive theories of decision making under uncertainty with emphasis on the ways in which human decisions depart from normative models of rationality; analytical foundations stemming from several disciplines, economics, psychology, management science; application in engineering systems will be considered.
- **DAEN 460 Capstone Senior Design (Spring 2026)**
Credits 3. 1 Lecture Hour. 6 Lab Hours. Second course in a two-course sequence for the capstone senior design experience; continuation of work on the senior design project in teams; data collection, analysis, application of data engineering methods and tools and development of recommendations considering design constraints, evaluation of alternative design, and application of relevant standards; engagement in oral presentations and creation of the project report, with relevant feedback provided during the semester.
- **DAEN 459 Capstone Senior Design Planning (Fall 2025)**
Credits 3. 2 Lecture Hours. 3 Lab Hours. First in a two-course sequence for the capstone senior design experience; formation of a senior design team, visitation with the team sponsor, preparation of the groundwork for the project, preparation of the project charter and collection or acquisition of initial set of data; provision of instructions on different aspects of capstone design,

including ethics, design constraints, applicable standards, project management, report writing specifications and requirements, and oral and visual presentations.

Landscape Architecture and Urban Planning, Texas A&M University

(Average Teaching Evaluation 4.59/5)

- **URPN 326 / PLAN 626 Advanced GIS in Urban-Regional Planning** (graduate-level; Spring 2025)
Credits 3. 3 Lecture Hours. Advanced instruction in applications of spatial tools for urban planning, landscape architecture, land development, hazard management, and related problems; GIS applications through review of literature and practice; data quality, uncertainty, the integration of GPS, remote sensing and information technology within the context of urban and regional planning.
- **URPN 325 Introduction to GIS in Urban and Regional Planning** (Fall 2024, Spring 2025)
Credits 3. 2 Lecture Hours. 3 Lab Hours. Provides an understanding of GIS fundamentals; basic concepts, principles and functions; essential skills for applying GIS in various fields such as urban planning, landscape architecture, land development, environmental studies, transportation and hazard management; based on learning through class projects.
- **URPN 310 Urban Analytical Methods II** (Fall 2024, Spring 2025)
Credits 3. 3 Lecture Hours. Focuses on research conducted by planners, sociologists, anthropologists, political scientists and a variety of applied social scientists; examines variety of procedures employed when conducting research in urban areas; furthers understanding and knowledge of statistical methods employed in social research and elements of geographical analysis.
- **URPN 210 Urban Analytical Methods I** (Fall and Spring 2019, 2020, 2021, 2022, 2023, 2024)
Credits 3. 3 Lecture Hours. Study of various analytical techniques used in urban and regional decision making; quantitative approaches to analyze and manipulate data; utilization of statistical packages for data, analysis and communication to enhance urban planning modeling.
- **PLAN 604 Planning Methods I** (graduate-level; Fall 2023, 2024)
Credits 3. 3 Lecture Hours. Fundamental concepts and methods used in urban and regional research; qualitative and quantitative research designs; measurement and scaling; sampling; data collection; data file construction; introduction to data analysis and statistical inference.
- **URSC 642 Analytic Methods in Landscape and Urban Research II** (graduate-level; Spring 2024)
Credits 3. 3 Lecture Hours. Provides a survey of hands on experiences with advanced techniques and procedures related to conceptual measurement and operational issues, data set development and manipulation and data analysis issues critical for conducting academic research.
- **URPN 201 Evolving Cities** (Fall 2021, Spring 2024)
Credits 3. 3 Lecture Hours. Introduction to the history of contemporary urban and regional planning and how the evolving forms of cities and regions pose opportunities and/or challenges for planners; understanding key social, economic, political and technological forces that shape city form and function and its ramification for urban and regional planning.
- **FYEX 101 First Year Experience** (Fall 2022)
Credits 0. 0 Lecture Hours. Development of self-efficacy, self-awareness and a sense of purpose; active engagement in the learning environment inside and outside of the classroom; social integration within the university community; also taught at Galveston and Qatar campuses. Must

be taken on a satisfactory/unsatisfactory basis.

4.2 Guest Lecturer

- **URSC 631 Foundations of Planning**, presenting *Natural Hazards Research* (Spring 2025)
Department of Landscape Architecture and Urban Planning, Texas A&M University
Instructors: Sinan Zhong, Ji Won Nam
- **URPN 361 Urban Issues**, presenting *GIS in Planning Communities* (Fall 2022)
Department of Landscape Architecture and Urban Planning, Texas A&M University
Instructor: Madhavi Dixit
- **PLAN 610 Structure and Function of Urban Settlements**, presenting *Urban Sprawl* (Fall 2016)
Department of Landscape Architecture and Urban Planning, Texas A&M University
Instructor: Han Park

4.3 Teaching Assistant

- **URSC 642 Analytic Methods II** (Spring 2017, 2018, 2019)
Department of Landscape Architecture and Urban Planning, Texas A&M University
Instructor: Walt G. Peacock
- **URPN 201 Evolving Cities** (Spring 2017)
Department of Landscape Architecture and Urban Planning, Texas A&M University
Instructor: Yu Xiao
- **Coding, Introduction to Production Engineering** (Spring 2006)
Department of Production Engineering, Federal University of São Carlos, Brazil
Instructor: Reinaldo Morabito, Vitoria Pureza
- **Operations Research Applied to Management** (Fall 2005)
Department of Production Engineering, Federal University of São Carlos, Brazil
Instructor: Reinaldo Morabito
- **Operations Research Applied to Logistics** (Fall 2005)
Department of Production Engineering, Federal University of São Carlos, Brazil
Instructor: Reinaldo Morabito

5 Research

5.1 Peer Reviewed Journals

1. Blanks, J., Abuabara, A., Roberts, A., Semien, J. (2021). Preservation at the Intersections: Patterns of Disproportionate Multihazard Risk and Vulnerability in Louisiana's Historic African American Cemeteries. *Environmental Justice*, 14 (1), 1-14.
2. Horney, J.A., Karaye, I.M., Abuabara, A., Gearhart, S., Grabich, S., Perez-Patron, M. (2020). The Impact of Natural Disasters on Suicide in the United States. *Crisis*, 42 (5), 328-334.
3. Abuabara, A., Abuabara, A., Tonchuk, C. (2017). Comparative Analysis of Death by Suicide in Brazil and in the United States: descriptive, cross-sectional time series study. *Sao Paulo Medical Journal*, 135, 150-156.
4. Abuabara, A., Morabito, R. (2009). Cutting Optimization of Structural Tubes to Build Agricultural Light Aircrafts. *Annals of Operations Research*, 169 (1), 149-165.
5. Abuabara, A., Morabito, R. (2008). Modelos de Programação Inteira Mista para o Planejamento do

Corte Unidimensional de Tubos Metálicos na Indústria Aeronáutica Agrícola. *Gestão & Produção*, 15 (3), 605-617.

5.2 Research Projects Collaboration

1. *Safeguarding Texas's Inland Rural Heritage: Collaborative Strategies for Resilience Against Natural Hazards* (Spring/Fall 2026), funded by the COA Small Grant Program and the Center for Heritage Conservation, co-sponsorship by the Melbern G. Glasscock Center for Humanities Research, Texas A&M University. Pls: Michelle A. Meyer, Fabrizio Aimar, Alexander Abuabara, John T. Cooper Jr., Seth H. Jordan.
2. *Hurricane Evacuation Studies*: (i) [Southeastern Texas \(2024-2025\)](#), (ii) [Texas Coastal Bend \(2018-2019\)](#), and (iii) [Texas Valley \(2014-2016\)](#), funded by the US Army Corps of Engineers (USACE) and the Federal Emergency Management Agency (FEMA). Co-Pls at TAMU: Walt G. Peacock, J. Andy Mullins (*in memoriam, Andy passed away during the course of this project, his vision and leadership continue to guide our work*), David H. Bierling, Darrell W. Borchardt, Douglas F. Wunneburger, Alexander Abuabara.
3. *Risk-Based Community Resilience Planning* (2022-2024), funded by Colorado State University from the National Institute of Standards and Technology (NIST). Directors: John W. van de Lindt (CSU), Jamie Brown Kruse (CSU); Co-Pls at TAMU: Walt G. Peacock, Shannon S. Van Zandt, Nathanael P. Rosenheim, Michelle A. Meyer, Maria Koliou.
4. *The Adoption and Utilization of Hazard Mitigation Practices by Jurisdictions along Gulf and Atlantic Coasts* (2016-2018), funded by the National Science Foundation. Co-Pls: Walt G. Peacock, Shannon S. Van Zandt, Himanshu Grover (SUNY).

5.3 Reports

1. [*in preparation*] Bierling, D.H., Wunneburger, D.F., Peacock, W.G., Borchardt, D.W., Abuabara, A. (2025). Southeast Texas Hurricane Evacuation Study: Transportation and Vulnerability Report. Texas A&M Hazard Reduction & Recovery Center and Texas A&M Transportation Institute. College Station, TX.
2. [*in preparation*] Bierling, D.H., Wunneburger, D.F., Peacock, W.G., Borchardt, D.W., Abuabara, A. (2025). Southeast Texas Hurricane Evacuation Study: Evacuation and Sheltering Behavior Report. Texas A&M Hazard Reduction & Recovery Center and Texas A&M Transportation Institute. College Station, TX.
3. Lee, R.J., Day, W., Abuabara, A., Newman, G.D., Peacock, W.G. (2023). The Effect of Flood Buyouts on Nearby Property Values. Working paper for the *Lincoln Institute of Land Policy*.
4. Rosenheim, N., Lane, G., Katare, A., Watson, M., Williams, A., Peacock, W.G., Abuabara, A., Perez, M., Sullivan, E., Kastor, H. (2021). Food Retailer Data, in *Food Access Impact Survey for Harris County and Southeast Texas after Hurricane Harvey in 2017*.
5. Peacock, W.G., Abuabara, A., Wunneburger, D.F., Bierling, D.H., Mullins, J.A., Borchardt, D.W. (2020). Coastal Bend Hurricane Evacuation Study: Evacuation Zone Development Report. Texas A&M Hazard Reduction & Recovery Center and Texas A&M Transportation Institute. College Station, TX.
6. Peacock, W.G., Wunneburger, D.F., Abuabara, A., Bierling, D.H., Mullins, J.A., Borchardt, D.W. (2020). Coastal Bend Hurricane Evacuation Study: Vulnerability Analysis Report. Texas A&M Hazard Reduction & Recovery Center and Texas A&M Transportation Institute. College Station, TX.

7. Bierling, D.H., Lindell, M., Peacock, W.G., Abuabara, A., Moore, R., Wunneburger, D.F., Mullins, J. A., Borchardt, D.W. (2020). Coastal Bend Hurricane Evacuation Study: Hurricane Harvey Evacuation Behavior Survey Outcomes and Findings. Texas A&M Hazard Reduction & Recovery Center and Texas A&M Transportation Institute. College Station, TX.
8. Mullins, J.A., Borchardt, D.W., Bierling, D.H., Peacock, W.G., Wunneburger, D.F., Abuabara, A. (2020). Coastal Bend Hurricane Evacuation Study: Transportation Analysis Report. Texas A&M Hazard Reduction & Recovery Center and Texas A&M Transportation Institute. College Station, TX.
9. Peacock, W.G., Wunneburger, D.F., Abuabara, A., Park, H., Van Zandt, S. (2016). Rio Grande Valley Hurricane Evacuation Study: Vulnerability Analysis Report. Texas A&M Hazard Reduction & Recovery Center and Texas A&M Transportation Institute. College Station, TX.

5.4 Data

1. Lane, G., Rosenheim, N., Williams, A., Watson, M., Abuabara, A., Maher, D., Rathburn, A., Peacock, W.G. (2024). Food Aid Agency Data, in Food Access Impact Survey for Harris County and Southeast Texas after Hurricane Harvey in 2017. DesignSafe-CI.

5.5 Conferences with Peer Reviewed Abstracts

(underlined name indicates the presenter)

1. Wunneburger, D.F., Peacock, W.G., Abuabara, A., Bierling, D.H., Borchardt, D.W. Risk-Based Hurricane Evacuation Zone Planning for the Texas Gulf Coast. *2025 Researchers Meeting, organized by the International Sociological Association Research Committee on Disasters (ISA-RC39)*. Broomfield, CO (Jul 2025)
2. Stanley, M., Peacock, W.G., Abuabara, A., Day, W., Wade, H. Procedural and Distributive Equity in Disaster Housing Recovery: FEMA's IA Program for Unmet Housing Needs After Hurricane Ike *50th Annual Natural Hazards Research and Applications Workshop*. Broomfield, CO (Jul 2025)
3. Day, W., Abuabara, A., Peacock, W.G., Wade, H. FEMA's Financial Housing Assistance to Owners and Renters After Hurricane Harvey. *49th Annual Natural Hazards Research and Applications Workshop*. Broomfield, CO (Jul 2024)
4. Blanks, J., Abuabara, A. Did You Check on the Cemetery? A Geospatial Risk Assessment of Cemeteries in Southeast Louisiana. *PRIMR Conference at PVAMU*. Prairie View, TX (Mar 2023)
5. Abuabara, A., Day, W., Peacock, W.G., Wade, H. An Analysis of FEMA's Individual Assistance After Hurricane Harvey. *Natural Hazards Research Summit*. Washington DC (Oct 2022)
6. Abuabara, A., Peacock, W.G., Bierling, D.H., Lindell, M. A Statistical Analysis of Household Evacuation in Response to Hurricane Harvey. *Society for Risk Analysis Annual Meeting*. Austin, TX (Dec 2020)
7. Abuabara, A., Medina-Cetina, Z., Peacock, W.G. Application of Bayesian Networks to Hurricane Risk Assessment. *Society for Risk Analysis Annual Meeting*. Arlington, VA (Dec 2019)
8. Horney, J.A., Karaye, I.M., Gearhart, S., Abuabara, A., Perez-Patron, M. Impact of Natural Disasters on Suicide Risk in the United States. *American Public Health Association Annual Meeting*. Philadelphia, PA (Nov 2019)
9. Abuabara, A., Peacock, W.G. Importance of Mobile Homes to Determine Natural Hazard Evacuation Boundaries. *46th Annual Natural Hazards Research and Applications Workshop*. Broomfield, CO (Jul 2017)

10. Abuabara, A., Considerations for Modeling Natural Hazard Evacuation of Mobile Neighborhoods and Mobile Parks in the Texas Valley Region. *42nd Urban Affairs Assoc. Conference*. San Diego, CA (Mar 2016)
11. Abuabara, A., Morabito, R. Modelo de Otimização de Corte Unidimensional Aplicado a Fabricação de Aeronaves Agrícolas, *Annals of XXXIX Simpósio Brasileiro de Pesquisa Operacional*. Fortaleza, Brazil (Aug 2007)
12. Morabito, R., Abuabara, A. A One-dimensional Cutting Stock Problem in the Production of Agricultural Light Aircrafts", *XXII European Conference on Operational Research*, Prague, Czech Republic (Jul 2007)
13. Abuabara, A., Morabito, R. Pipe Cutting Optimization: Application to the Aeronautics Industry. *Latin-Iberoamerican Conference on Operations Research CLAIO*. Montevideo, Uruguay (Nov 2006)

5.6 Workshops, Presentations, Invited Talks, Seminars & Panel Participation

1. Designing for Impact: Faculty Perspectives on High-Impact and Reflective Learning (Panelist). Texas A&M Center for Teaching Excellence. College Station, TX (Jan 27, 2026). Organizer: Nate Poling
2. Introduction to QGIS. Texas A&M GIS Day. College Station, TX (2018, 2019, 2020, 2021, 2025) Organizer: Daniel Goldberg et al.
3. Problem Structuring, Data, and My Teaching Journey, and Why My Students Still Ask for Extra Credit. ISEN Spring Seminar. College Station, TX (2025)
4. Risk-based Evacuation Zone Planning for the Texas Gulf Coast. CoA Research Symposium. College Station, TX (Fall 2025)
5. Zone Planning for Coastal Texas. CoA Research Symposium. College Station, TX (2024)
6. Application of Bayesian Networks to Hurricane Risk Assessment. CoA Research Symposium. College Station, TX (Fall 2024)
7. Modeling Individual Assistance to Households in Texas After Hurricane Harvey. CoA Research Symposium. College Station, TX (Fall 2024)
8. Hurricane Evacuation Studies: Zone Planning for Coastal Texas. CoA Research Symposium. College Station, TX (Fall 2024)
9. Modeling Individual Assistance to Households in Texas After Hurricane Harvey. CoA Research Symposium. College Station, TX (Fall 2023)
10. A Web-Based GIS Platform and Applications for Evacuation, Preparedness, Response, Mitigation, and Recovery Planning: The Texas Hazard Resilience Atlases. Texas Emergency Management Conference. College Station and Dallas, TX (Spring 2023)
11. FEMA Individual Assistance Program for Housing. IN-CORE Summer Research Meeting. Fort Collins, CO (Spring 2022)
12. Spatial Analysis in R. Texas A&M GIS Day. College Station, TX (2018, 2019, 2020, 2021) Organizer: Daniel Goldberg et al.
13. Hurricane Evacuation Studies. Texas A&M GEOSAT Seminar. College Station, TX (Fall 2018)

6 Service

1. Undergraduate Affairs Committee (Fall 2025-present)

Department of Industrial and Systems Engineering, Texas A&M University
Chair: Amarnath Banerjee

2. Faculty Senator (Fall 2024-Summer 2025)
College of Architecture, Texas A&M University
COA Faculty Senator Caucus Leader: Shelley D. Holliday
Faculty Senate Speakers: Angie Hill and Andrew Klein
3. Faculty Member: Master of Urban Planning (MUP) reaccreditation, strategic planning (2024-2025)
Department of Landscape Architecture and Urban Planning, Texas A&M University
Coordinator: Ivis Garcia Zambrana
4. Faculty Member: Bachelor of Urban/Regional Planning accreditation, strategic planning (2024-2025)
Department of Landscape Architecture and Urban Planning, Texas A&M University
Coordinator: Justin Golbabai

7 Students

7.1 Current Students

External Committee Member

1. Adam Mann, Master of Urban Planning (Expected Spring 2026)
2. Aaron Garcia, Master of Urban Planning (Expected Spring 2026)

7.2 Capstone Faculty Advisor

Industrial & Systems Engineering, Texas A&M University

- 1.
- 2.
- 3.
4. Jan Barrett, Eric Gomez, Annemarie Hoffzimmer, Thomas Lim, Isha Pokerna, Gabrielle Wibner.
Pedestrian Conflict at Railroad Crossings. Sponsor: Cyndi Lee, Program Aide for the East End District, Houston, TX. *3rd place in major of the 2025 COE Project Showcase* (Fall 2025)

8 Honors, Fellowships, Awards

8.1 Professional Development

1. Certificate in Promoting Active Learning, Association of College and University Educators (ACUE, Fall 2025)
2. Faculty Affairs Third-Party Conflict Resolution Mediator, Texas A&M University (Fall 2024)
Associate Provost for Faculty Affairs: Heather Wilkinson
3. Graduate Mentoring Academy Fellow, Texas A&M University (Fall 2024)
Assistant Director of Mentoring & Graduate Education: Clinton A. Patterson

8.2 Professional Affiliations and Memberships

1. Faculty fellow for the Transforming Teaching through Engaged and Active Learning (TEAL) Academy (2025-present)
2. Faculty fellow, Hazard Reduction and Recovery Center, Texas A&M University (2015-present)
3. Society for Risk Analysis (2018-present)

4. Tau Sigma Delta Honor Society (2015-present)
5. Regional Council of Engineering and Agronomy of São Paulo, Brazil (2002-2005)

8.3 Honors and Fellowships

1. Travel Award for Natural Hazards Research Summit, National Science Foundation, Washington DC (Oct 2022)
2. Summer Institute for Early-Career Faculty Award, National Science Foundation Natural Hazards Engineering Research Infrastructure (NHERI), San Antonio, TX (Jun 2022)
3. Open-Source GIS Developers Summer School, GeoSTAT Institute of Botany of the Czech Academy of Sciences (IBOT), Prague, Czech (Aug 2018)
4. Coastal Flood Risk Reduction Research, National Science Foundation Partnerships for International Research and Education (PIRE), Amsterdam, Netherlands (May 2018)
5. Full Tuition Assistantship, Texas A&M University (2015-2022)
6. Graduate Program Scholarship, Department of Landscape Architecture and Urban Planning, Texas A&M University (2014-2015)
7. Graduate Program Scholarship, Brazilian Federal Agency for Support and Evaluation of Graduate Education (CAPES), Department of Production Engineering, Federal University of São Carlos, Brazil (2005-2006)
8. REHIDRO Monitoring Work, Undergraduate Scholarship, University of São Paulo, Brazil (1997-1998)
9. CISC Monitoring Work, Undergraduate Scholarship, University of São Paulo, Brazil (1994-1997)